

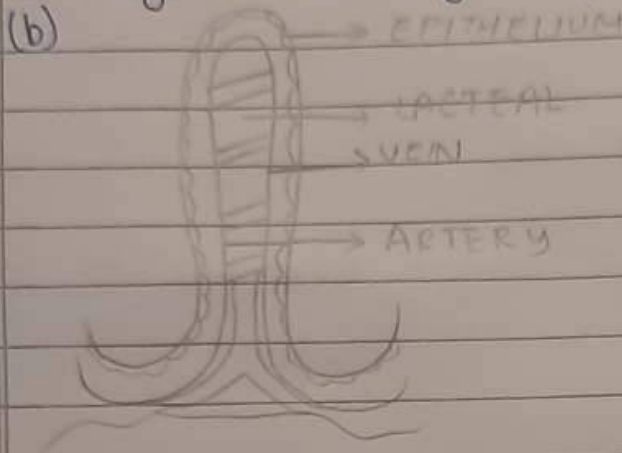
BIOLOGY

1. ATP and NADPH are the connecting link between light and dark reactions.
2. The process of aerobic respiration in eukaryotic cells occurs in the mitochondria.
3. The lymph is devoid of RBC's or leucocytes.
4. The plants use two major processes to get rid of excess water :
 - Transpiration.
 - Guttation.
5. Malpighian body comprises of glomerulus and Bowman's Capsule.
6. The age of plant does not affect rate of transpiration.
7. Plants do not breathe, they only respire through the stomata present in their leaves.
 - CO_2 released is not utilised by humans however it is utilised by plants when they do photosynthesis.
8.

LIGHT REACTION	DARK REACTION
• Occurs in grana part of chloroplast	• Occurs in stroma region of plant
• No sugar formation takes place	• Sugar formation takes place.
• Oxygen is released	• Oxygen isn't released.
9. (a) Pepsinogen and prorennin are the enzymes that act on proteins.

(b) When there is absence of oxygen the 3-carbon molecule of carbon turns into Ethanol, CO_2 and energy.

10. (a) The role of HCl in the stomach is to activate the enzymes that digest proteins.



The main function of lacteal is to absorb large molecules of fats and lipids from the small intestine.

11. (a) The alveoli of lungs are covered with blood capillaries so that exchange of gases takes place between alveoli and surrounding capillaries.

(b) The cartilaginous ring is very strong and flexible. It supports the trachea and prevents it from falling.

12. (a) In winter, less water is used for metabolism and thus there is less concentration of ADH and more concentration of DCT. As there is more DCT, lesser reabsorption of H_2O occurs and more urine is formed.

(b) The skin, liver, lungs, salivary glands and large intestine are the accessory excretory organs.

(c) The arthropoda have malpighian tubes (bowman's capsule and glomerulus)

13. (i) 4 - left Atricle

5 - ~~inferior venacava~~ tricuspid Valve

(ii) 3 is superior venacava and its main function is to get deoxygenated blood from the upper part of body.

6 is inferior venacava and its main function is to get deoxygenated blood from the lower part of body.

(iii) The blood vessels that carry ~~de~~ away blood from the heart are:

• Arteries. → Ex: Aorta

(iv) The circulation in hearts of human is called double circulation as the blood flows through the heart twice per circuit.